Lab Day 3

Quiz 1

Your Task:

Hospital Management System

- Doctor (Doctor_id, Specialization, Qualification, Name[First, Middle, Last])
- Patient (Patient_id, Age[derived], DOB, Address[Locality, Town/City])
- Medicine (Code, Price, Quantity)

Relationships:

- Doctor 1 — treats — n Patient => doctor m - m patient
- Patient m — bill — n Medicine

Quiz 1

First run this queries:

```
INSERT INTO Doctor (Doctor_id, Specialization, Qualification, Name_First, Name_Middle, Name_Last)
VALUES
(1, 'Cardiology', 'MD', 'Ahmed', 'Ali', 'Hassan'),
(2, 'Neurology', 'PhD', 'Mona', 'Mohamed', 'Fahmy'),
(3, 'Orthopedics', 'MBBS', 'Khaled', NULL, 'Saeed'),
(4, 'Pediatrics', 'MD', 'Sara', 'Ahmed', 'Nabil'),
(5, 'Dermatology', 'MD', 'Omar', 'Hassan', 'Farouk');
```

Quiz 1

```
INSERT INTO Patient (Patient_id, DOB, Address_Locality, Address_City)
VALUES
(101, '1990-05-12', 'Heliopolis', 'Cairo'),
(102, '1985-11-23', 'Zamalek', 'Cairo'),
(103, '2000-01-30', 'Nasr City', 'Cairo'),
(104, '2010-07-15', 'Maadi', 'Cairo'),
(105, '1975-09-10', 'Dokki', 'Giza');
```

Quiz 1

```
INSERT INTO Medicine (Code, Price, Quantity)
VALUES
('M001', 150.50, 20),
('M002', 75.00, 50),
('M003', 300.25, 10),
('M004', 120.00, 30),
('M005', 50.75, 100);
```

Quiz 1

```
-- Doctor 1 treats Patients 101, 102
INSERT INTO Doctor_Treats_Patient (Doctor_id, Patient_id)
VALUES
(1, 101),
(1, 102),
(2, 103),
(3, 104),
(4, 105),
(2, 101);
```

Quiz 1

```
INSERT INTO Patient_Bills_Medicine (Patient_id, Medicine_Code, Quantity)
VALUES
(101, 'M001', 2),
(101, 'M002', 1),
(102, 'M003', 1),
(103, 'M002', 3),
(104, 'M004', 2),
(105, 'M005', 5);
```

Quiz 1

1. A patient was added by mistake with patient_id = 5.

   Remove this patient from the system.

2. Display all doctors who work in Cardiology department and their salary is greater than 12000.

3. Display all patients whose name starts with the letter "M".

4. Show doctors whose salary is between 10000 and 20000.

5. Display doctors who specialize in Cardiology or Dermatology.

6. Display doctors who do not work in the Neurology department.

Quiz 1

7. Find all patients who did not enter their phone number.

8. Display doctor name and salary, and create a new column called salary_status:

If salary > 15000 → "High Salary"

Otherwise → "Normal Salary"

9. Display all patients who have appointments with doctor_id = 1.

10. Create a new table called high_salary_doctors and insert doctors whose salary is greater than 15000.

11. Display the doctors who have at least one appointment with a patient in the system.

Quiz 1

12. Display doctors whose salary is greater than the salary of at least one doctor in the Cardiology department.

13. SIMILAR TO / Regex

Display all patients whose name starts with "A" or "M" using pattern matching.

14. DISTINCT

Show a list of all unique doctor specializations.

15. Calculated Columns

Display patient name and age (calculated from dob assuming today’s date).

Quiz 1

16.UPPER / LOWER / INITCAP

Display all patient names in uppercase, lowercase, and capitalized format (first letter of each word uppercase).

17.TRIM / LTRIM / RTRIM

Clean up the phone column by removing leading and trailing spaces.

18.CONCAT

Display full patient info combining name and phone as a single column called contact_info.

19.SUBSTRING / POSITION

Extract the first 3 letters of patient names and find the position of letter "a" in the name.

Quiz 1

20.REPLACE

In doctor names, replace "Ahmed" with "Ahmad".

21.CASTING

Display doctor salary as integer and as text in separate columns.

Assignment

First run this queries:

```
INSERT INTO Department (dept_id, dept_name, location) VALUES
(1, 'IT', 'New York'),
(2, 'HR', 'London'),
(3, 'Finance', 'Tokyo'),
(4, 'Marketing', 'Paris'),
(5, 'Operations', 'Dubai');
```

Assignment

First run this queries:

```
INSERT INTO Employee (emp_id, emp_name, salary, hire_date, dept_id) VALUES
(101, 'Alice Johnson', 7500, '2022-01-15', 1),
(102, 'Bob Smith', 5000, '2021-03-20', 2),
(103, 'Charlie Brown', 9000, '2020-07-10', 1),
(104, 'David Lee', 3500, '2023-05-01', 3),
(105, 'Eva Green', 12000, '2019-11-25', 4),
(106, 'Frank White', 4000, '2022-08-12', NULL),
(107, 'Grace Kim', 6000, '2021-12-30', 5);
```

Assignment

First run this queries:

```
INSERT INTO Project (project_id, project_name, budget) VALUES
(201, 'Project Alpha', 50000),
(202, 'Project Beta', 75000),
(203, 'Project Gamma', 100000),
(204, 'Project Delta', 25000),
(205, 'Project Epsilon', 80000);
```

Assignment

First run this queries:

```
INSERT INTO Works_On (emp_id, project_id, hours) VALUES
(101, 201, 120),
(101, 202, 80),
(102, 203, 150),
(103, 201, 100),
(104, 204, 90),
(105, 205, 200),
(107, 202, 50);
```

Assignment

Display all employees who work in the IT department

Display employees whose salary is BETWEEN 4000 AND 9000

Display employees whose name starts with 'A' using SIMILAR TO (PostgreSQL Regex Operators)

Display employees who do NOT belong to any department

Display employees who work in departments 1, 2, or 3 using IN

Assignment

Display employee name and salary level using CASE WHEN

High if salary > 8000

Medium if salary BETWEEN 4000 AND 8000

Low if salary < 4000

Display employees who are assigned to at least one project using EXISTS

Display employees whose salary is greater than ANY salary in department 2

Display the employee who has the HIGHEST salary using a subquery

Create a table called High_Salary_Employees and insert employees whose salary > 8000 using INSERT INTO … SELECT

Assignment

Delete employees who are NOT assigned to any project

Display departments that do NOT have any employees using NOT EXISTS

Display all DISTINCT department locations

Create a calculated column showing salary after 10% bonus

Display employee names in UPPER, LOWER, and INITCAP formats

Remove spaces from employee names using TRIM, LTRIM, and RTRIM

Concatenate first_name and last_name into a single column using CONCAT

Assignment

Extract a substring of employee names using SUBSTRING

Find the POSITION of a character or substring in employee names using POSITION

Replace a part of employee name using REPLACE

Change the data type of salary to INTEGER using CAST